Improving Buyer/Seller Transparency in the Secondary Market for Trading Card Collectibles

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MIS 312 - 1 Intro to System Analysis and Design

Preliminary Study

Problem

Selling trading cards as an individual is time consuming due to the nature of how buying and selling works. Individuals must go through and find valuable cards by manually typing in a card's name on to an online marketplace.

Objective

Our goal is to create a mobile app that will automatically scan a card and show the user the price. This will cut down on the time it takes for sellers to sort and search for valuable cards.

Scope

The initial scope will encompass creating a basic prototype that will allow for users to take a picture and have a pre trained machine learning model detect the card automatically. It will then reference online databases to find the card's price and display it. Our initial sample will be Pokemon cards.

Plan for Project

Our plans for this project is to utilize machine learning image recognition to detect trading cards and cross referencing online marketplaces for prices.

Feasibility Study

- Explain each section max-5 lines
- Technical Feasibility: We will be using a graphical neural network to identify the cards corresponding to a database of all cards
- Economic Feasibility : There is sufficient demand for this sort of app, however at this time we are unsure of how we would market the app to compensate for costs
- Operational Feasibility: User support will depend on how successful our program identifies cards
- Value Analysis

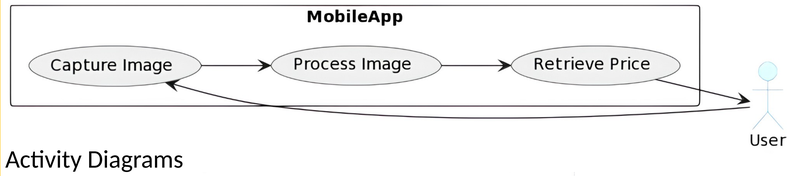
System Analysis -1

Explain each section max-5 lines

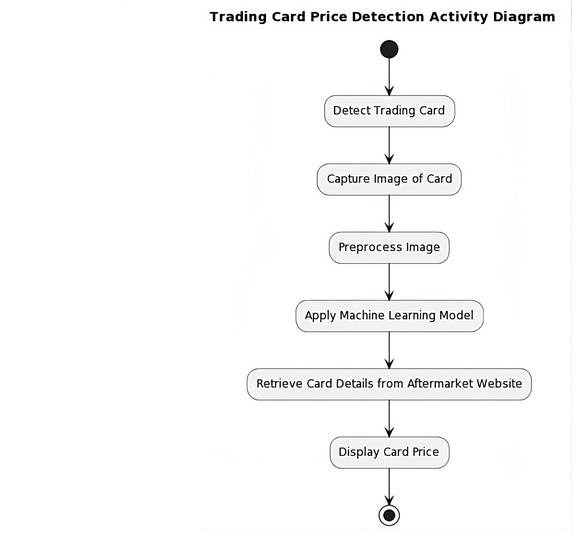
- Users take a photo of their card and upload it, where it is then identified and then our database sends back the corresponding card and its associated value
- The only existing way for people to perform an operation like this is by hand, manually identifying cards yourself and listing them online.
- Currently we have identified one data source through kaggle.com that contains a large set of pokemon cards. Other data sources will need to be added if we wanted to expand our scope

System Analysis - 2

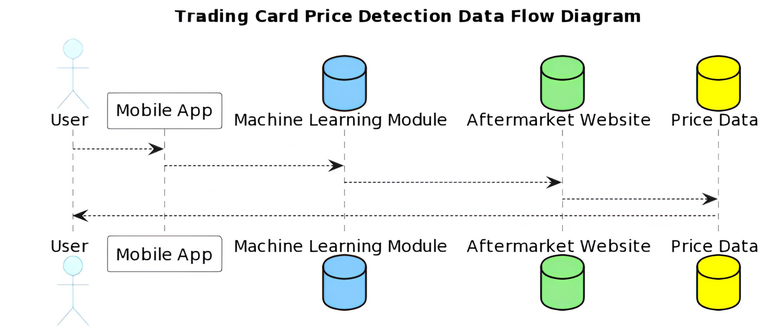
- Use Case Diagrams



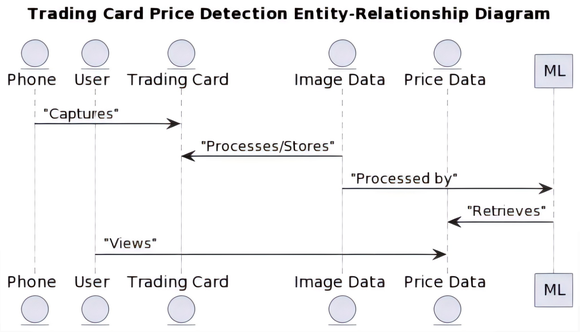
Activity Diagrams



- Data Flow Diagram

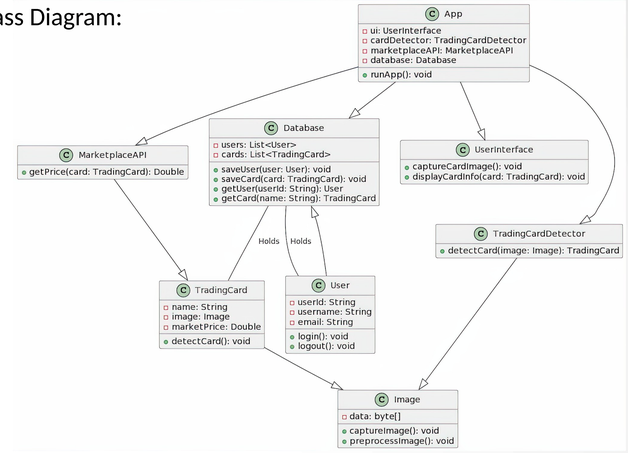


- Entity-Relationship (E-R) Diagram

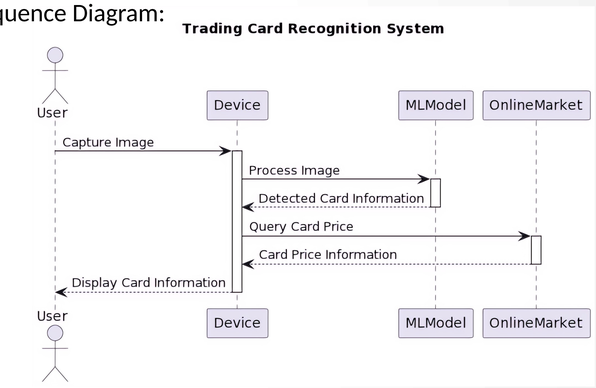


Results

- UML Class Diagram:



- UML Sequence Diagram:



- Design Patterns

Project Overview

- Our project is a smart phone app whose purpose is to automatically detect trading cards using the device's camera and referencing online aftermarket marketplaces to display to users the price of their scanned trading card.

Conclusion

The project offers a notable improvement in the trading card sales process. Utilizing machine learning for image recognition and integrating with online marketplaces, the app is designed to ease the process of determining prices for trading cards, potentially saving time for sellers and money for buyers.

Explain each section max-5 lines